

27.4.20

Tutorial - 4

*1.

$$v_{gs} = v_a \sin \omega t, \quad v_{ds} = v_{DD}$$

- Triode limit :- $v_{DD} - v_a > v_{in} - v_T$
 Assuming
 negligible
 swing at i/x
 $v_{x, min} > v_{sat} = 0$
 $\Rightarrow v_a \approx v_{DD}$

- Cutoff limit :- $\frac{v_a}{\alpha_{in}} = I_b$ (I_b - bias current)

$$a) \quad \eta_d = \frac{\rho_{out}}{\rho_d} = \frac{(v_a/\sqrt{2})^2}{\alpha_{in}} \times \frac{1}{v_{DD} \times I_b} = \frac{v_a}{2 v_{DD}}$$

$$\therefore \eta_d \approx \frac{1}{2} = 50\%$$

$$b) \quad \phi_{M_1} = v_x \times I_x = (v_{DD} + v_a \sin \omega t) \left(I_b - \frac{v_a \sin \omega t}{\alpha_{in}} \right)$$

$$\approx (v_{DD} + v_{DD} \sin \omega t) (I_b - I_b \sin \omega t)$$

$$\phi_{M_1} = v_{DD} I_b \times (1 - \sin^2 \omega t) = v_{DD} I_b \times \cos^2 \omega t$$

$$\therefore \rho_{M_1, avg} = \langle \phi_{M_1} \rangle = v_{DD} I_b \langle \cos^2 \omega t \rangle = \frac{1}{2} v_{DD} I_b$$

$$\therefore \rho_{M_1, avg} = \frac{1}{2} \times \rho_{v_{DD}}$$

A 2. $v_y = -\frac{g_{m1}}{g_{m2}} v_{in}$

→ Assuming high gain (negligible swing at i/x and y)

$$v_y > v_{in} - V_T$$

$$\therefore v_{ds} - V_{GS2}/I_{ds} > v_{in} - V_T$$

$$\therefore v_{ds} > v_{in} + V_{DSAT2}$$

$$v_x > v_{ds} - V_T$$

$$V_{DD} - v_a > v_{ds} - V_T$$

$$v_a < V_{DD} - v_{ds} + V_T$$

$$\therefore v_{a, \max} = V_{DD} - (v_{in} + V_{DSAT2}) + V_T$$

$$= V_{DD} - V_{DSAT2} - v_{in} + V_T$$

$$= V_{DD} - V_{DSAT2} - V_{GS1}/I_{ds} + V_T$$

$$v_{a, \max} = V_{DD} - V_{DSAT2} - V_{DSAT1}$$

$$\eta_d = \frac{P_{out}}{P_{VDD}} = \frac{v_a^2}{2 \alpha_{in}} \times \frac{1}{V_{DD} I_{ds}}$$

$$\eta_d = \frac{(V_{DD} - V_{DSAT1} - V_{DSAT2})^2}{2 (I_{ds} \alpha_{in}) V_{DD}}$$

If $I_{ds} \alpha_{in} = V_a$:- $\eta_d = \frac{V_{DD} - V_{DSAT1} - V_{DSAT2}}{2 V_{DD}}$

$$\therefore \eta_d = \frac{1}{2} - \frac{(V_{DSAT1} + V_{DSAT2})}{2 V_{DD}}$$